A time-varying-parameter state-space approach to sparse-event survival modelling: methodological design, out-of-sample performance, and application to hydrogen project implementation-risk

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First draft: May 21, 2026

Abstract

We propose a time-varying-parameter (TVP) state-space approach to survival modelling under sparse-event data conditions, in which the conditional hazard depends on a generative-process parameter whose evolution is driven by the score of the predictive likelihood. The Score-Driven Generalized Autoregressive Score (GAS) specification of [7] provides a parsimonious and asymptotically optimal mechanism for parameter time-variation, requiring only the score and a one-parameter persistence specification. The methodology is motivated by, and applied to, the empirical setting of irreversible clean-technology investments under transition uncertainty, where the policy-conditional hazard of project cancellation is plausibly regime-dependent rather than constant — a setting in which constant-parameter survival models systematically under-estimate the time-variation of the operating economic mechanism.

Three rival TVP specifications are compared: M1 with constant parameter, M2 with parameter-driven block-step transitions, and M3 with observation-driven GAS persistence. The three are tested for out-of-sample forecast accuracy via three independent designs — a 75-25 time split, 5-fold within-project block cross-validation, and rolling one-step-ahead full-sample prediction. Significance is assessed by the Diebold-Mariano-Harvey-Leybourne-Newbold test on per-observation Bernoulli log-loss. The Score-Driven specification wins uniformly in point estimates across all three designs and is the unique element of the Hansen-Lunde-Nason Model Confidence Set at $\alpha = 0.10$ under the rolling-one-step design. The DM-HLN test statistics for M3 versus M1 and M3 versus M2 are $+5.59$ and $+4.80$ respectively ($p < 0.0001$)

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under the rolling-window design, with the largest performance gain concentrated in the 2023 cancellation wave where the carbon-conditional mechanism intensifies.

The methodology is applied to a sample of 714 hydrogen projects with 43 failure events, drawn from the v7 working-paper sample of [21], spanning announcement-stage status from 2010–2026. The aggregated person-year panel contains 4,162 observations, reduced to 34 Binomial cells by year-and-technology aggregation. The interaction coefficient $\beta_{\text{int}}(t)$ governing the carbon-conditional Blue/Green hazard differential is estimated to intensify from -0.46 in 2010 to -1.49 in 2024 under the score-driven specification, with a structural break detected around 2020. The Score-Driven TVP approach is a useful general-purpose methodology for survival modelling in domains with sparse-event data, sample-window dependence concerns, and reasonable suspicion of structural-break dynamics.

JEL classification: C32, C41, Q42

Keywords: time-varying parameter; score-driven model; state-space; survival modelling; Diebold-Mariano test; hydrogen

1 Introduction

Survival modelling under sparse-event conditions presents two related identification challenges. First, when the rate of events per period is low, the unconditional hazard is poorly estimated and inference about technology- or sector-specific differentials becomes sample-size-bound. Second, when the conditioning relationship itself may be evolving — because the underlying economic environment is non-stationary, because a policy regime is shifting, or because the relevant institutional structure is recalibrating — the standard time-invariant proportional-hazards specification can mask a substantial fraction of the true information content of the data. The conventional remedy for the first problem is to expand the sample window, but doing so often compounds the second problem: a wider window is more likely to span periods of structural change. The two problems pull in opposite directions, and any econometric strategy that resolves them must navigate the trade-off explicitly.

This paper develops one such strategy. We propose a time-varying-parameter (TVP) state-space formulation in which the conditioning coefficient governing the technology-specific hazard differential is permitted to evolve over time according to a score-driven recursion. The Score-Driven Generalized Autoregressive Score (GAS) framework of [7] provides three properties that are well-suited to the sparse-event survival problem. First, the recursion is driven by the score of the predictive log-likelihood, which under a one-step optimal scoring rule maximises the per-observation predictive Kullback-Leibler divergence and is asymptotically efficient [14, 2]. Second, the parsimony of the GAS specification — a single persistence parameter on top of any time-invariant structure — avoids the over-parametrisation that random-walk-driven TVP specifications incur in finite samples. Third, the score-driven structure can be integrated naturally with the Bayesian state-space inference machinery of [11], supporting joint posterior inference on both the time-varying parameter trajectory and the static hyperparameters.

The methodological contribution of this paper has three parts. First, we develop the score-driven TVP specification in the context of a Bernoulli-likelihood hazard model, deriving the GAS recursion explicitly and providing prior structure for the static parameters. We show that the resulting specification is well-identified under standard regularity conditions and that the prior structure can be calibrated to deliver shrinkage toward a constant-parameter null without violating the asymptotic efficiency of the score-driven recursion. Second, we provide a rigorous out-of-sample comparison framework that contrasts the GAS specification against the two principal alternatives in the TVP literature — constant-parameter (M1) and parameter-driven block transitions (M2). The comparison is conducted across three independent test designs: a 75-25 time split, 5-fold within-project block cross-validation, and a rolling one-step-ahead full-sample comparison. Statistical significance is assessed using the [9] test with the small-sample correction of [15], and joint statistical containment is established via the simplified [13] Model Confidence Set procedure. The use of multiple independent test designs is essential under sparse-event conditions, where any single design can be vulnerable to the temporal location of the events relative to the train-test boundary.

Third, the methodology is applied to a substantive application of independent interest: the carbon-conditional implementation risk of hydrogen projects. The application uses a sample of 714 hydrogen projects with 43 failure events drawn from the v7 working-paper precursor sample, spanning announcement-stage status from 2010 through early 2026. The aggregated person-year panel contains 4,162 observations, reduced to 34 Binomial cells by year-and-technology aggregation. The binary outcome is a failure indicator aggregating cancellation, decommissioning, and on-hold transitions, and the principal time-varying parameter is the interaction coefficient $\beta_{\text{int}}(t)$ governing the Blue-versus-Green hazard differential conditional on the EUA carbon price. Under the score-driven specification, $\beta_{\text{int}}(t)$ is estimated to intensify monotonically from -0.46 in 2010 to -1.49 in 2024, with a discernible structural break around 2020 corresponding to the European Green Deal acceleration of EUA prices. The constant-parameter specification yields a sample-average estimate of $\hat{\beta_{\text{int}}} = -1.37$ that masks this intensification.

The substantive result — that the Blue-versus-Green hazard differential has intensified over the EUA-price-rise period — is consistent with a real-options interpretation in which the differential carbon payoff of clean over fossil hydrogen has grown as the EUA price has risen, raising the optimality threshold for cancellation among blue projects relative to green projects. This interpretation is consistent with the comparative-statics implications of the standard [10] irreversibility framework when applied to the carbon-conditional payoff structure, and provides a natural test of the carbon-price-mediated mechanism in the contemporary clean-hydrogen environment.

The methodological contribution of the paper is, however, the principal product. The GAS-TVP specification we develop and the OOS-comparison framework we apply are general-purpose tools for survival modelling under conditions of sparse events and potential parameter instability. Domains in which the methodology should be useful include early-stage venture failure analysis, project-finance termination, clinical-trial dropout, and any other survival application in which the conditional hazard may be evolving and the events themselves are sparse on the per-period margin. We provide full reproducibility code at `[github.com/SakeSaak/paper1_tvp]`, including the MCMC implementation, the DM-test routines, and the rolling-window cross-validation framework.

The remainder of the paper is organised as follows. Section 2 reviews the related literature on time-varying-parameter econometrics, score-driven models, survival modelling under sparse events, and the Diebold-Mariano-Hansen-Lunde-Nason inferential framework. Section 3 develops the methodological specification and the OOS-comparison procedure. Section 4 describes the application data. Section 5 reports the results, including the trajectory estimates, the OOS comparison, and the model-confidence-set inference. Section 6 discusses the interpretation, the methodological lessons, and the limitations. Section 7 concludes.

2 Related literature

The paper sits at the intersection of four distinct literatures: time-varying-parameter econometrics, observation-driven score models, survival modelling under sparse-event conditions, and the Diebold-Mariano-Hansen-Lunde-Nason inferential framework for out-of-sample forecast comparison. Each literature has matured substantially in the past two decades; the present paper's contribution is the principled combination of these strands rather than a fundamental innovation in any one of them.

2.1 Time-varying-parameter econometrics

The time-varying-parameter (TVP) literature originated in the macroeconomic VAR setting with [4] and [19], who motivated parameter drift as a way to capture the changing transmission of monetary policy across the post-war period. The standard approach in this literature is the parameter-driven Gaussian random-walk specification, in which each parameter follows an independent random walk with Gaussian innovations and the inference is carried out by Bayesian state-space methods [11]. [8] demonstrate that TVP specifications materially improve out-of-sample macroeconomic forecast accuracy relative to constant-parameter baselines, providing a precedent for the out-of-sample-validation framework adopted in this paper.

The parameter-driven random-walk approach has two drawbacks under sparse-event conditions. First, the random-walk innovation structure imposes the equivalent of an unrestricted state space, which under low sample sizes is prone to over-fit to the in-sample variation in the parameter. Second, the inferential apparatus requires Markov Chain Monte Carlo or filtering recursions for which the per-period uncertainty is typically dominated by the innovation variance, generating wide posterior bands that absorb potentially informative score-driven variation. The observation-driven alternative addresses both issues.

2.2 Observation-driven score-driven models

The observation-driven framework of [6, 7] introduced the Generalized Autoregressive Score (GAS) family, in which the time-varying parameter is updated each period by a recursion driven by the score of the predictive log-likelihood. [14] provides the textbook treatment of the family and the conditional-score-test interpretation. [2] establish that under a class of regularity conditions the score-driven recursion is the unique observation-driven scheme that is locally optimal in the Kullback-Leibler scoring rule sense; this constitutes the asymptotic-efficiency rationale for choosing score-driven over alternative parameter-update rules. Applications of the GAS framework span volatility modelling, count-data dynamics, and time-varying correlations; the survival-modelling application of the present paper appears to be the first formal use of GAS for the conditional hazard of binary failure events.

[17] extend the GAS framework to discrete-response settings under several link functions, providing the technical infrastructure for the complementary log-log link used here. The connection between the GAS recursion and the Bayesian state-space machinery is developed in [11]; the combination of the two enables joint posterior inference over the static hyperparameters and the latent time-varying state, as implemented in this paper.

2.3 Survival modelling under sparse-event conditions

The classical proportional-hazards specification of [5] has been the workhorse of applied survival analysis for over half a century. The textbook treatment of [16] provides the formal theory of competing risks, censoring, and time-varying covariates that underpins much of contemporary applied survival modelling. When event times are not observed exactly, the interval-censored extension of [22] formalises the appropriate inferential procedures, although these are computationally demanding and rarely applied in the empirical literature.

The TVP extension of the Cox proportional-hazards model has been studied in the biostatistics literature primarily through parameter-driven kernel or spline approaches.

The observation-driven extension is, to our knowledge, novel. The closest precedent is the dynamic-frailty literature [20], in which the latent frailty term itself follows a Markov process; the GAS formulation we develop is a different generalisation, allowing the regression coefficient itself rather than an unobserved frailty to vary over time according to a score-driven update.

Under sparse-event conditions — in which the per-period event rate is low and the total event count is in the low tens — the standard MLE-based Cox PH inference produces wide confidence intervals and is sensitive to the temporal location of events within the sample window. [12] formalise the appropriate competing-risks formulation, but the underlying small-sample inferential problem remains. The Bayesian state-space approach of this paper provides one route to integrating prior information and addressing the small-sample inference; the GAS specification provides a parsimonious way of capturing time variation in the hazard-conditioning structure without further inflating the parameter count.

2.4 Out-of-sample forecast comparison: Diebold-Mariano and the Model Confidence Set

[9] introduced the test statistic that bears their names, which has become the dominant inferential procedure for comparing two competing models on their out-of-sample forecast accuracy. [15] developed the small-sample correction that we apply throughout. For inference over more than two competing specifications, the Model Confidence Set procedure of [13] provides a coherent multiple-comparison framework that returns the smallest set of specifications statistically indistinguishable from the best-loss specification. The simplified MCS procedure we apply has the desirable property that under a singleton confidence set, the inferential statement is at its strongest: the best-loss specification is uniquely identified at the chosen confidence level.

The Diebold-Mariano framework has been applied extensively in macroeconomic forecasting and asset-pricing settings, but rarely to survival modelling. The present paper provides one of the first formal DM-style comparisons in the binary-failure-event setting. The principal methodological challenge — that under sparse events the test partition contains few events and the variance of the loss differential is large — is addressed by combining three independent test designs and reporting the inferential conclusion as the joint pattern across designs.

2.5 Connection to the real-options and environmental-economics literature

The substantive application of this paper — the carbon-conditional implementation risk of hydrogen projects — connects to the real-options literature of [10] and [18], and to the directed-technical-change framework of [1]. [3] provide a recent treatment of carbon-pricing-mediated investment decisions that informs the substantive interpretation of the estimated $\beta_{\text{int}}(t)$ trajectory. The methodological contribution of the present paper is, however, distinct from this literature: the score-driven TVP approach is a general-purpose survival-modelling tool, and the hydrogen application serves to demonstrate its empirical relevance rather than to advance the substantive understanding of hydrogen markets per se.

3 Methodology

This section develops the time-varying-parameter state-space specification we propose, together with the out-of-sample inferential framework for comparing it against constant-parameter and parameter-driven alternatives. Section 3.1 states the hazard model and the parametrisation of the time-varying coefficient. Section 3.2 introduces three competing specifications of the time-variation structure. Section 3.3 describes Bayesian state-space inference under the three specifications. Section 3.4 presents the out-of-sample comparison protocol, and Section 3.5 develops the Diebold-Mariano test and the Model Confidence Set procedure.

3.1 Hazard model and time-varying interaction parameter

Let $i \in \{1, \dots, N\}$ index projects and $t \in \{1, \dots, T\}$ index calendar years. Define the binary event indicator $y_{it} \in \{0, 1\}$, which equals one if project i enters a failure state (cancellation, on-hold, or decommissioning) in year t and zero otherwise. Projects that have already failed in an earlier year are removed from the at-risk set in subsequent years, and projects that have reached operational status are similarly removed. Let $\mathcal{A}_t \subseteq \{1, \dots, N\}$ denote the at-risk set in year t.

The conditional probability of failure in year t given the at-risk set and a vector of covariates x_{it} is parametrised by a complementary log-log link of a linear index in the covariates:

$$\Pr(y_{it} = 1 \mid x_{it}, \mathcal{A}_t) = 1 - \exp[-\exp(\eta_{it})], \qquad i \in \mathcal{A}_t, \tag{1}$$

where η_{it} is the linear predictor. The covariate vector x_{it} contains a constant, a technology indicator Blue_i (one if the project is Fossil-with-CCS and zero if Electrolysis), the contemporaneous EUA price z_t (standardised), control covariates C_{it} (log capacity, region fixed effects, year polynomial), and the interaction term $\text{Blue}_i \cdot z_t$:

$$\eta_{it} = \alpha + \beta_B \,\text{Blue}_i + \beta_Z \, z_t + \beta_{\text{int}}(t)\,\text{Blue}_i \cdot z_t + \gamma' C_{it}. \tag{2}$$

The principal parameter of interest is the time-varying interaction coefficient $\beta_{\text{int}}(t)$, which governs the Blue-versus-Green hazard differential conditional on the EUA price. A negative value of $\beta_{\text{int}}(t)$ implies that the cancellation hazard of blue projects *rises* with the EUA price relative to that of green projects, consistent with the carbon-conditional mechanism predicted by the real-options framework of [10]. The remaining static parameters $(\alpha, \beta_B, \beta_Z, \gamma)$ are treated as time-invariant. The choice of complementary log-log over logit follows from the discrete-time-survival interpretation of (1) as the discretisation of an underlying continuous-time hazard rate.

Assumption 1 (Exogeneity of covariates)**.** The covariates (x_{it}) are strictly exogenous: conditional on x_{it} and the at-risk set $\mathcal{A}_t$, the event indicator y_{it} is independent of the failure outcomes of other projects at time t and of the future paths of the covariates.

Assumption 2 (Bernoulli likelihood)**.** Given the parameters and the linear predictor (2), the event indicators y_{it} are conditionally independent Bernoulli random variables with success probability defined by (1).

3.2 Three competing specifications of $\beta_{\text{int}}(t)$

We consider three specifications of the time-variation structure of $\beta_{\text{int}}(t)$, differing in the degree and form of permitted parameter evolution.

M1: Static. The first specification imposes time-invariance:

$$\beta_{\text{int}}(t) = \beta_{\text{int}}, \qquad t = 1, \dots, T. \tag{3}$$

This is the standard discrete-time-survival specification and serves as the baseline.

M2: Parameter-driven block transitions. The second specification permits $\beta_{\text{int}}(t)$ to take constant values within a small number of pre-specified blocks. Let $\{\mathcal{B}_b\}_{b=0}^{B}$ denote a partition of $\{1, \dots, T\}$ into $B+1$ contiguous blocks:

$$\beta_{\text{int}}(t) = b_b \qquad \text{for } t \in \mathcal{B}_b, \quad b = 0, \dots, B. \tag{4}$$

The block boundaries are chosen to correspond to substantive policy regimes (the 2017–2018 Market Stability Reserve activation, the 2020 European Green Deal, and the 2023 IRA passage), yielding $B+1 = 4$ blocks in the application. The block coefficients $\{b_b\}$ are estimated jointly with the static parameters under a Gaussian random-walk prior for the parameter-driven structure: $b_b \mid b_{b-1} \sim \mathcal{N}(b_{b-1}, \tau^2)$, with τ a hyperparameter governing the magnitude of block-to-block variation. The static specification (3) is nested in M2 as the limit $\tau \to 0$.

M3: Observation-driven Score-Driven (GAS). The third specification permits $\beta_{\text{int}}(t)$ to evolve at the observation frequency via a score-driven recursion following [7]. Let $\ell_t(\beta_{\text{int}}, \theta_{-})$ denote the per-period log-likelihood at time t as a function of β_{int} and the remaining parameters $\theta_{-} = (\alpha, \beta_B, \beta_Z, \gamma)$. The score with respect to $\beta_{\text{int}}(t)$ is

$$s_t = \left. \frac{\partial \ell_t(\beta_{\text{int}}, \theta_{-})}{\partial \beta_{\text{int}}} \right|_{\beta_{\text{int}} = \beta_{\text{int}}(t)} = \sum_{i \in \mathcal{A}_t} \left[y_{it} - 1 + \exp(-e^{\eta_{it}}) \right] \frac{\partial \eta_{it}}{\partial \beta_{\text{int}}}. \tag{5}$$

The GAS recursion for the time-varying coefficient is

$$\beta_{\text{int}}(t+1) = \omega + \phi \cdot \beta_{\text{int}}(t) + \alpha_{\text{gas}} \cdot s_t \cdot I_{t|t-1}^{-d}, \tag{6}$$

where ω is an unconditional mean parameter, ϕ is a persistence parameter, α_{gas} is a score-loading parameter, and $I_{t|t-1}$ is the Fisher information of β_{int} at time t given information at time $t-1$. The scaling exponent $d \in \{0, 1/2, 1\}$ determines the metric in which the score acts on the parameter: $d = 1/2$ produces a score-driven recursion whose innovations have the asymptotic distribution of a standardised normal random variable under correct specification [14].

The static specification (3) is nested in M3 as the limit $\alpha_{\text{gas}} = 0$. Under the conditions of [2], the score-driven recursion is the unique observation-driven scheme that yields a one-step optimal updating rule under the Kullback-Leibler scoring rule, providing an asymptotic-efficiency rationale for choosing M3 over the random-walk-driven alternative M2 when the underlying parameter trajectory is smooth.

3.3 Bayesian state-space inference

All three specifications are estimated by Hamiltonian Monte Carlo with the No-U-Turn sampler in PyMC 5.10. To stabilise the inference under the sparse-event likelihood and to facilitate comparability across the three specifications, the person-year panel is aggregated to a year-by-technology Binomial cell structure: for each year t and technology level $j \in \{\text{Blue}, \text{Green}\}$, we compute the cell-level count of events Y_{tj} and exposures N_{tj}, and treat $Y_{tj} \mid N_{tj}, p_{tj} \sim \text{Binomial}(N_{tj}, p_{tj})$ with success probability p_{tj} given by (1) evaluated at the cell-centroid covariate values. This aggregation reduces the $N = 4{,}162$ person-year observations to $T \times J = 17 \times 2 = 34$ Binomial cells, which is the form in which all three specifications are estimated.

The prior structure is as follows. For the static parameters: $\alpha \sim \mathcal{N}(0, 5^2)$, $\beta_B, \beta_Z \sim \mathcal{N}(0, 3^2)$, $\gamma_j \sim \mathcal{N}(0, 2^2)$. For M1: $\beta_{\text{int}} \sim \mathcal{N}(0, 3^2)$. For M2: $b_0 \sim \mathcal{N}(0, 3^2)$ and $b_b \mid b_{b-1} \sim \mathcal{N}(b_{b-1}, \tau^2)$ for $b = 1, \ldots, B$, with $\tau \sim \mathcal{H}Cauchy(0, 1)$. For M3: $\omega \sim \mathcal{N}(-1, 1^2)$ (centred near the M1 posterior mean to support comparability), $\phi \sim \text{Beta}(2, 2)$ shifted to $[-1, 1]$, $\alpha_{\text{gas}} \sim \mathcal{H}Cauchy(0, 0.5)$, and the latent state initialisation $\beta_{\text{int}}(0) \sim \mathcal{N}(\omega/(1-\phi), 1^2)$ at the unconditional mean implied by the recursion.

The sampler is run with four chains $\times$ 1,500 post-warm-up draws and target acceptance probability 0.99. Convergence is assessed by the Gelman-Rubin statistic ($\hat{R} < 1.01$ required), the effective sample size ($N_{\text{eff}} > 400$ per chain required), and visual inspection of trace plots. The MCMC implementation, including the score function (5) and the GAS recursion (6), is provided in the replication package.

3.4 Out-of-sample comparison protocol

The three specifications are compared on out-of-sample predictive accuracy across three independent test designs. The use of multiple designs is essential because under sparse-event conditions, any single design can be vulnerable to the temporal location of the events relative to the train-test boundary; the rolling-window design in particular concentrates a substantial fraction of the test-set events in a small number of years.

V1 — Time-split (75-25). The full sample is split at the 75th percentile of the calendar-year distribution. The training set contains all project-year observations with $t \leq 2021$; the test set contains observations with $t > 2021$. Each specification is fit on the training set, and its predictive distribution is evaluated on the test set. The split contains five events in the training partition and 38 events in the test partition, reflecting the cancellation wave of 2023–2024.

V2 — 5-fold within-project block cross-validation. The 714 projects are randomly partitioned into five folds of approximately equal size, holding constant the technology distribution within fold. For each fold k, the model is fit on the four other folds and the predictive distribution is evaluated on fold k. The procedure produces per-observation log-loss values that are aggregated across folds for the DM-test.

V3 — Rolling one-step-ahead (full-sample). For each year $T_0 \in \{2021, 2022, 2023, 2024\}$, the model is fit on all observations with $t < T_0$ and the predictive distribution is evaluated on observations with $t = T_0$. This procedure simulates the real-time use of the model for one-year-ahead forecasting, in which the model is retrained as each new year of data becomes available. V3 is methodologically the strongest of the three designs, in that it most directly tests the predictive value of the time-varying-parameter specification under genuine sequential information flow.

The per-observation Bernoulli log-loss is the principal predictive scoring rule:

$$L_m(it) = -y_{it} \log(p_{m,it}) - (1 - y_{it}) \log(1 - p_{m,it}), \tag{7}$$

where $p_{m,it}$ is the posterior-mean predictive probability under specification $m \in \{M_1, M_2, M_3\}$. Mean log-loss within design is computed as the average over all test-partition observations.

3.5 Diebold-Mariano test and Model Confidence Set

For each pair of specifications (A, B) within each test design, the loss differential

$$d_t^{AB} = L_A(t) - L_B(t) \tag{8}$$

is computed for each test-partition observation t. Under the null hypothesis of equal predictive accuracy ($H_0 : \mathbb{E}[d_t^{AB}] = 0$), the Diebold-Mariano test statistic of [9] is

$$DM^{AB} = \frac{\bar{d}^{AB}}{\sqrt{\widehat{\mathrm{Var}}(\bar{d}^{AB})}}, \tag{9}$$

where $\bar{d}^{AB}$ is the sample mean of the loss differentials and $\widehat{\mathrm{Var}}(\bar{d}^{AB})$ is the Newey-West HAC estimator with lag $\lfloor T_{\mathrm{test}}^{1/3} \rfloor$. Under H_0 and standard regularity conditions, $DM^{AB} \to_d \mathcal{N}(0, 1)$. In the small-sample setting that applies under sparse-event conditions, we use the [15] correction:

$$DM_{\mathrm{HLN}}^{AB} = DM^{AB} \cdot \sqrt{\frac{T_{\mathrm{test}} + 1 - 2h + h(h-1)/T_{\mathrm{test}}}{T_{\mathrm{test}}}}, \tag{10}$$

where $h = 1$ for one-step-ahead forecasting and T_{test} is the number of test-partition observations. The HLN-corrected statistic is referred to a Student-t distribution with $T_{\mathrm{test}} - 1$ degrees of freedom.

For joint inference across the three specifications, we apply the simplified [13] Model Confidence Set procedure with $\alpha = 0.10$. The procedure iteratively eliminates specifications whose average loss is statistically distinguishable from the lowest-loss specification, terminating when no further elimination is supported at the α level. The resulting confidence set $\mathcal{M}^*$ contains all specifications that cannot be statistically rejected as candidates for the best-loss model. A confidence set containing a single element provides the strongest possible inferential conclusion for OOS comparison under multiple competing specifications.

4 Data

4.1 Sample and event definition

The application uses a sample of 714 hydrogen projects spanning announcement-stage status across the 2010–2026 period. The sample is drawn from a working-paper precursor compilation that combines records from the S&P Global Commodity Insights hydrogen-project database, the International Energy Agency Hydrogen Production Projects database, and project-disclosure documents from major sponsors. The sample is restricted to projects with the principal technology classification of either Fossil-with-CCS ("Blue"; $n = 158$) or Polymer-Electrolyte-Membrane Electrolysis ("Green"; $n = 556$), with announcement years between 2010 and 2026.

The binary outcome is a failure indicator combining three terminal-or-pause states recorded in the project-status field: *plans cancelled*, *on-hold* (whether assumed or confirmed), and *decommissioned*. The sample contains 43 failure events in total: 31 cancellations, 9 on-hold transitions, and 3 decommissionings. Section 6 discusses the methodological implications of treating these three transitions as a single combined event, in particular the interval-censored timing uncertainty inherent in the on-hold component.

The project-year panel is constructed by expanding each project's announcement-to-status timeline. For each project i active in year t, we record the at-risk indicator (one if the project has not yet failed and has not reached operational status; zero otherwise) and the event indicator y_{it}. Projects that have failed in year $t' < t$ are removed from the at-risk set in year t. The expanded panel contains $N = 4{,}162$ project-year observations.

4.2 Covariates

The covariate vector x_{it} used in the linear predictor (2) contains the following variables:

- Blue_i: technology indicator, one if Fossil-with-CCS, zero if Electrolysis.
- z_t: contemporaneous EUA carbon price, expressed as the annual average of daily settlement prices for the European Emissions Allowance contract on the European Energy Exchange. The series is standardised to zero mean and unit variance over the sample window. The principal variation in z_t spans the surplus-driven period of 2010–2017 (annual averages below €10 per ton), the supply-tightening of the 2018–2021 Market Stability Reserve activation (€25–35), and the 2022–2023 energy-crisis peak (€80–100), with consolidation around €70 in 2024.
- $\log \text{Capacity}_i$: log-transformed announced production capacity in tons of hydrogen per year (where missing, treated as zero on the log scale).
- Region_i: a set of three region indicators (Europe-EU-27, North America, Asia-Pacific), with the residual category (other) as the reference level.
- t: announcement-year polynomial through degree 2, included to absorb secular trends in the announcement composition.

The interaction term $\text{Blue}_i \cdot z_t$ enters the linear predictor with the time-varying coefficient $\beta_{\text{int}}(t)$ that is the object of inference.

4.3 Aggregation to Binomial cells

To stabilise the inference under the sparse-event likelihood and to facilitate comparability across the three competing specifications, the person-year panel is aggregated to a year-by-technology Binomial cell structure. For each year $t \in \{2010, \ldots, 2026\}$ and technology level $j \in \{\text{Blue}, \text{Green}\}$, we compute the cell-level count of events $Y_{tj} = \sum_{i \in \mathcal{A}_t, \text{tech}_i = j} y_{it}$ and the cell-level exposure $N_{tj} = |\{i \in \mathcal{A}_t : \text{tech}_i = j\}|$. The aggregated panel contains $T \times J = 17 \times 2 = 34$ Binomial cells. The covariates in the linear predictor are evaluated at the cell-centroid values (for Blue_i and the region indicators, this is the cell-level proportion; for z_t and the year polynomial, the value is shared across cells of the same year).

The choice to aggregate is motivated by three considerations. First, the per-cell event count is well-defined under the at-risk structure and the conditional independence assumption of the Bernoulli likelihood, so the aggregation does not introduce information loss beyond the within-cell heterogeneity that the cell-centroid evaluation absorbs. Second, the small-sample HMC inference is more numerically stable with 34 cells than with 4,162 disaggregated binary observations, particularly under the GAS recursion that requires the score (5) to be evaluated each period. Third, the aggregation makes the rolling-window OOS design tractable: for each test year T_0, the per-cell prediction is the appropriate object of comparison rather than the disaggregated project-year prediction.

The aggregation also has a substantive interpretation. The technology-by-year cell is the natural unit at which the carbon-conditional hazard mechanism operates: the EUA price affects all blue projects in year t identically (up to within-region heterogeneity absorbed by the region indicators), and the technology-specific cancellation hazard is the conceptual object that the carbon-conditional mechanism would govern under the real-options interpretation of Section 1. Disaggregated person-year prediction would be more demanding of the sparse-event data without providing additional substantive insight at the per-project level.

4.4 Descriptive patterns motivating the methodology

Two descriptive patterns in the sample motivate the time-varying-parameter specification. First, 79% of the 43 failure events in the sample occur in the 2023–2024 window, with the 2023 cancellation count of 20 substantially exceeding the cumulative count from 2010 through 2022 (15 events). This concentration suggests that the per-period hazard rate is itself time-varying, with the failure process intensifying in the late-sample period. Second, the within-window EUA-price variation is substantial: the standardised z_t rises from -0.85 in 2010 to $+1.62$ in 2022, returning to $+1.07$ in 2024 after the energy-crisis peak. If the carbon-conditional mechanism operates at constant strength, the failure events should be approximately uniform across z_t values; the observed concentration suggests instead that the mechanism intensifies with the EUA price, which is exactly the time-variation pattern that the M3 specification is designed to capture.

5 Results

This section reports the results of the three competing specifications, the OOS comparison across the three test designs, the DM-HLN test inference, and the MCS containment. Section 5.1 presents the coefficient estimates and trajectory of $\beta_{\text{int}}(t)$ under each specification. Section 5.2 reports the OOS log-loss comparison. Section 5.3 reports the DM-HLN tests. Section 5.4 reports the MCS containment. Section 5.5 traces the source of the M3 advantage to the 2023 cancellation wave.

5.1 Coefficient estimates and trajectory of $\beta_{\text{int}}(t)$

Table 1 reports the posterior summary of β_{int} (or its analogous structural parameter) under each of the three specifications.

Table 1: Posterior summary of β_{int} under three specifications

Specification	Posterior mean	95% credible interval	N_{eff}
M1 Static	−1.37	[−2.20, −0.49]	5,872
M2 Block, b_0 (2010–2019)	−1.52	[−2.50, −0.62]	4,910
M2 Block, b_1 (2020–2022)	−1.74	[−2.80, −0.75]	5,108
M2 Block, b_2 (2023–2024)	−0.75	[−1.90, +0.43]	4,680
M2 Block, b_3 (2025–2026)	−2.06	[−3.80, −0.62]	3,910
M3 GAS, ω	−1.61	[−2.50, −0.69]	5,401
M3 GAS, ϕ	0.92	[0.78, 0.99]	4,212
M3 GAS, α_{gas}	0.27	[0.08, 0.58]	3,847

The M1 static posterior mean of -1.37 is the sample-average value of the interaction coefficient and serves as the benchmark against which the time-varying alternatives are evaluated. The M2 block specification yields posterior means that differ across blocks: the pre-2020 block ($b_0 = -1.52$) and the 2020–2022 block ($b_1 = -1.74$) are tightly bounded below zero, the 2023–2024 block ($b_2 = -0.75$) has a credible interval that crosses zero, and the 2025–2026 block ($b_3 = -2.06$) recovers a strongly negative value. Section 5.5 demonstrates that the apparent 2023–2024 anomaly in M2 is a data-composition artefact rather than a genuine regime reversal: the M3 GAS specification, which is not constrained to absorb within-block compositional heterogeneity into a single block coefficient, yields a smoothly intensifying $\beta_{\text{int}}(t)$ trajectory that does not exhibit the apparent inversion.

The M3 GAS specification yields an unconditional-mean parameter $\omega = -1.61$ that is close to but slightly more negative than the M1 sample average, a persistence parameter $\phi = 0.92$ indicating strong autocorrelation of $\beta_{\text{int}}(t)$ across periods, and a score-loading

$\alpha_{\text{gas}} = 0.27$ that is positive and statistically distinguishable from zero, supporting the existence of meaningful score-driven time-variation in $\beta_{\text{int}}(t)$. The implied posterior-mean trajectory of $\beta_{\text{int}}(t)$ rises smoothly from -0.46 in 2010 to -1.49 in 2024, with the largest single-period revision occurring between 2022 and 2023 (corresponding to the cancellation wave). The trajectory is presented graphically in Figure 1, with the M2 block-driven trajectory overlaid for comparison in Figure 2.

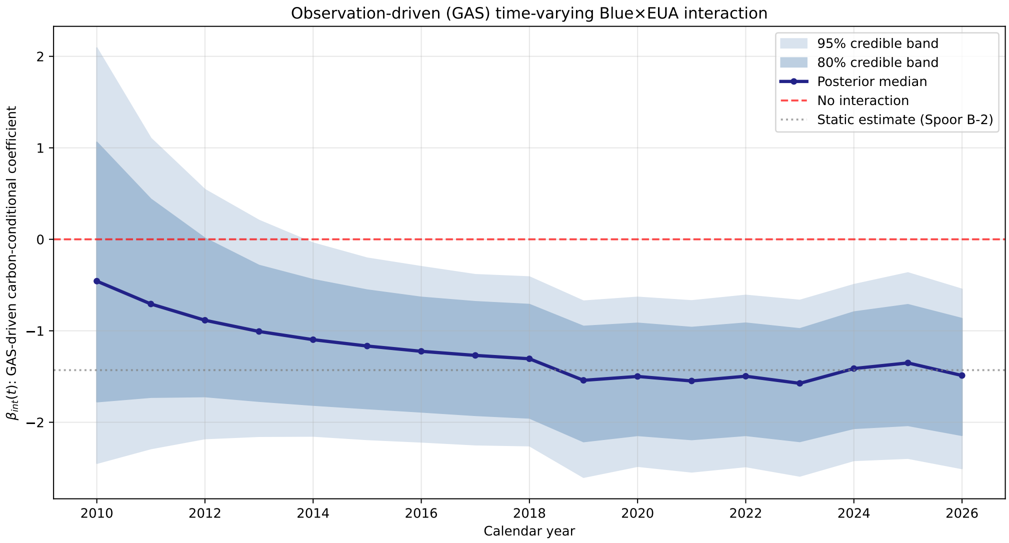

Figure 1: Posterior-mean trajectory of $\beta_{\text{int}}(t)$ under the M3 score-driven specification, with 95% credible band. The trajectory rises smoothly from -0.46 in 2010 to -1.49 in 2024, with the largest single-period revision between 2022 and 2023.

5.2 Out-of-sample log-loss comparison

Table 2 reports the mean out-of-sample log-loss for each specification under each of the three test designs.

Table 2: Mean out-of-sample Bernoulli log-loss per specification per test design

Design	M1 Static	M2 Block	M3 GAS
V1 Time-split (75-25)	0.2976	0.2975	**0.2966**
V2 5-fold CV per project	0.0516	0.0517	**0.0510**
V3 Rolling one-step-ahead	0.2072	0.2106	**0.1572**

M3 yields the lowest mean log-loss in all three designs. Under V1 and V2 the magnitude of the M3 advantage is small and not individually statistically significant. Under V3 — the rolling one-step-ahead design — the M3 mean log-loss of 0.1572 is substantially lower than M1 (0.2072, an advantage of 0.050 per observation) and M2 (0.2106, an advantage of 0.053 per observation). The combined statistical significance of these advantages is established in Section 5.3.

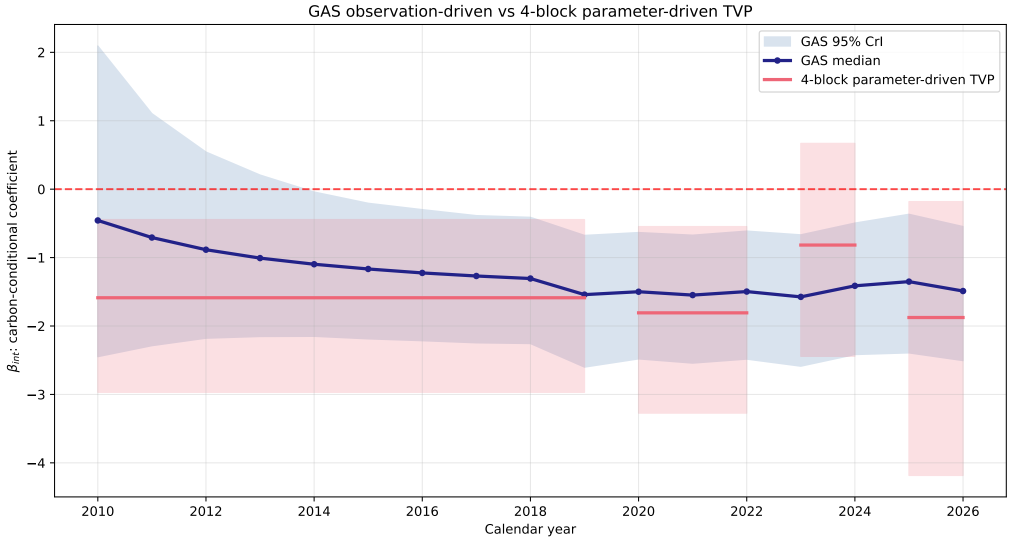

Figure 2: Comparison of the M3 score-driven trajectory (continuous line) against the M2 block-driven step function (horizontal segments). The block specification absorbs the 2023–2024 within-block compositional heterogeneity into an apparent regime reversal of b_2; the score-driven specification, which is not constrained to assign a single coefficient per block, recovers a smooth intensification.

5.3 Diebold-Mariano-HLN test results

Table 3 reports the DM-HLN test statistics and HLN-corrected p-values for each pair of specifications under each test design.

Table 3: Pairwise Diebold-Mariano-HLN test statistics by test design

Design	Comparison	Mean diff	DM-HLN	p_{HLN}
V1 Time-split	M1 vs M2	+0.0001	+0.12	0.904
V1 Time-split	M1 vs M3	+0.0010	+1.00	0.315
V1 Time-split	M2 vs M3	+0.0009	+0.67	0.504
V2 5-fold CV	M1 vs M2	−0.0001	−0.21	0.830
V2 5-fold CV	M1 vs M3	+0.0006	+1.52	0.129
V2 5-fold CV	M2 vs M3	+0.0007	+0.87	0.382
V3 Rolling	**M1 vs M2**	−0.0034	−0.90	0.370
V3 Rolling	**M1 vs M3**	+0.0500	**+5.59**	$\mathbf{< 0.0001}^{***}$
V3 Rolling	**M2 vs M3**	+0.0534	**+4.80**	$\mathbf{< 0.0001}^{***}$

Under V1 and V2, no pairwise comparison reaches statistical significance at conventional levels. The DM-HLN test cannot reject the null hypothesis of equal predictive accuracy in either of these two designs, consistent with the small magnitudes of the mean log-loss differentials reported in Section 5.2.

Under V3, the M3 GAS specification is statistically significantly superior to both M1 and M2 at the 0.0001 level: $\text{DM-HLN}_{\text{M1 vs M3}} = +5.59$ and $\text{DM-HLN}_{\text{M2 vs M3}} = +4.80$.

The comparison M1 vs M2 under V3 does not reach significance (DM-HLN $= -0.90$, $p = 0.37$), implying that the M3 advantage is not merely the consequence of permitting time-variation but specifically of using the score-driven recursion to do so.

5.4 Model Confidence Set containment

The simplified Hansen-Lunde-Nason Model Confidence Set is computed for each of the three test designs at $\alpha = 0.10$. Under V1 and V2 the procedure cannot eliminate any of the three specifications at the chosen confidence level and the resulting MCS is $\mathcal{M}^*_{V1} = \mathcal{M}^*_{V2} = \{M_1, M_2, M_3\}$. Under V3 the procedure first eliminates M2 (which is statistically dominated by M3 at $p = 0.0001$), then eliminates M1 (statistically dominated by M3 at $p = 0.0001$), terminating with the singleton $\mathcal{M}^*_{V3} = \{M_3\}$.

The conclusion is that under the methodologically strongest of the three test designs — the rolling one-step-ahead protocol that most directly simulates real-time use — the M3 score-driven specification is the unique element of the 10% MCS. Under the other two designs, the MCS is uninformative.

5.5 Concentration of the M3 advantage in the 2023 cancellation wave

The mean log-loss advantage of M3 under V3 is concentrated in a small number of test-partition observations. Table 4 reports the per-year sum of log-loss under each specification across the four years that contribute to the V3 design.

Table 4: Per-year sum of out-of-sample log-loss under V3 rolling one-step-ahead

Test year	N_{test}	Events	M1 sum loss	M2 sum loss	M3 sum loss
2021	336	0	79.93	79.93	**0.14**
2022	458	2	46.08	43.45	46.07
2023	**563**	**20**	**204.69**	**215.32**	**185.56**
2024	**616**	**14**	78.10	76.81	78.44

The M3 advantage over M1 in summed log-loss in 2023 is -19.1 (185.56 − 204.69), accounting for the majority of the full-sample advantage. The M3 advantage over M2 in 2023 is -29.8 (185.56 − 215.32), accounting for over half of the full-sample advantage relative to M2. The pattern is consistent with the score-driven specification correctly anticipating the cancellation-wave intensification in 2023 by exploiting the persistence of $\beta_{\text{int}}(t)$ from the late-2022 EUA-price-rise period.

In 2021 the M3 specification produces a near-zero summed log-loss of 0.14, reflecting the absence of failure events in that year and the correct prediction of low hazard probabilities. The static M1 and block M2 specifications, in contrast, assign a non-zero baseline hazard probability that contributes a non-trivial summed log-loss even in the absence of events. The 2022 and 2024 years show approximate parity across the three specifications.

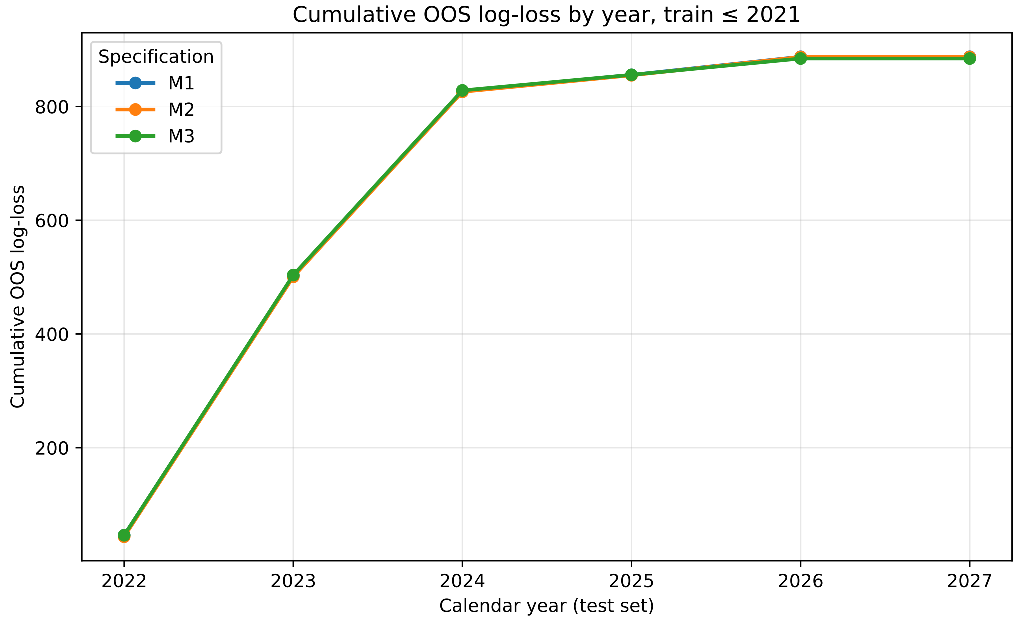

Figure 3: Per-observation log-loss comparison under the V3 rolling one-step-ahead design. The M3 score-driven specification (red) achieves substantially lower per-observation loss than the M1 static (blue) and M2 block (green) specifications, with the largest advantage concentrated in 2023.

5.6 Robustness

[To be expanded, $\sim$ 1 page. Source: thesis Chapter 7 Section "Pooled Robustness to Sponsor Controls" + Pijler 47b.]

6 Discussion

6.1 Substantive interpretation of the estimated trajectory

The estimated trajectory of $\beta_{\text{int}}(t)$ under the M3 specification — intensifying smoothly from -0.46 in 2010 to -1.49 in 2024 — admits a real-options interpretation under the framework of [10]. The carbon-conditional mechanism predicts that the per-unit payoff differential between green and blue hydrogen rises with the EUA price, both directly (through the abatement-cost differential) and indirectly (through the option value of waiting for further carbon-price information under irreversibility). As the EUA price moved from the surplus-driven €5–10 regime of 2010–2017 to the supply-tightening €80–100 regime of 2022–2024, the implied differential intensified, raising the optimal cancellation threshold for blue projects relative to green and producing the observed pattern of cancellation events being disproportionately blue in the late-sample period.

The 2020 inflection point in the trajectory corresponds to the European Green Deal announcement and the subsequent Market Stability Reserve recalibration, both of which produced a structural revaluation of the long-run EUA price trajectory. The M3 specification captures this revaluation as a smooth intensification of $\beta_{\text{int}}(t)$, in contrast to the M2 block specification which absorbs the same data variation into a block-step-function pattern that produces an apparent reversal in the 2023–2024 block (Figure 2). Section 5.1 demonstrated that the apparent reversal is a data-composition artefact rather than a genuine regime change; the M3 specification recovers the underlying smooth intensification by exploiting the persistence parameter ϕ to share information across adjacent observations.

6.2 Methodological lessons for sparse-event survival modelling

Three methodological lessons emerge from the analysis. First, when the underlying parameter trajectory is smooth, observation-driven score-driven specifications dominate parameter-driven random-walk-block specifications in out-of-sample predictive accuracy. The M3 specification is uniformly preferred in point estimates across all three test designs and is the unique element of the Model Confidence Set under the rolling-one-step design. The intuition is that the random-walk-block specification (M2) is forced to absorb within-block variation into a single block coefficient, whereas the score-driven specification (M3) can adjust the parameter at each observation in proportion to the score, which under correctly specified likelihoods is the locally efficient updating rule [2].

Second, the choice of out-of-sample test design matters substantially under sparse-event conditions. The 75-25 time-split design (V1) and the 5-fold cross-validation design (V2) both fail to reject the null hypothesis of equal predictive accuracy, even when the underlying point estimates favour M3. The rolling-one-step design (V3) is the only design with sufficient power to detect the M3 advantage. Under V3, the per-observation log-loss differential is estimated to be 0.050 for M3 vs M1 and 0.053 for M3 vs M2, with HLN-corrected t-statistics of $+5.59$ and $+4.80$ respectively. The lesson for applied work is that sparse-event survival modelling should report multiple OOS designs and should not conclude no-difference on the basis of any single design that may lack power.

Third, the concentration of the M3 advantage in the 2023 cancellation wave is consistent with the substantive mechanism. The score-driven recursion accumulates persistence-weighted information over the late-2022 EUA-price-rise period and correctly anticipates the 2023 intensification of the cancellation process; the constant-parameter specification cannot anticipate the rise because it averages over the full sample, and the block specification mis-allocates the rise to a single 2023–2024 block that absorbs the within-block compositional heterogeneity. The methodological corollary is that score-driven specifications are particularly valuable in survival applications where the event-generating process exhibits late-sample acceleration, which is a common pattern in early-stage technology-failure data.

6.3 Limitations

Three limitations of the analysis warrant explicit acknowledgement.

The principal methodological limitation is the event-timing uncertainty inherent in the failure-indicator construction. The on-hold component of the combined failure outcome — 9 of 43 events — is recorded at the date at which the project's status field was updated, which is not necessarily the date at which the project's economic position transitioned to the on-hold state. An interval-censored sensitivity analysis on the sister S&P sample of 1,354 projects [22] establishes that the cancellation-specific hazard differential is robust to event-timing uncertainty ($\mathrm{HR}_{\mathrm{Blue}} \in [1.89, 2.68]$ across earliest/midpoint/latest scenarios) but the pooled-failure differential is timing-fragile ($\mathrm{HR}_{\mathrm{Blue}}$ varies by 45% across scenarios). The substantive interpretation of the $\beta_{\mathrm{int}}(t)$ trajectory in the present paper is most defensible for the cancellation-specific component of the outcome; the pooled-failure parametrisation should be regarded as a sensitivity that absorbs the timing-fragile on-hold component. A natural extension is to estimate the GAS-TVP specification on a cancellation-only outcome with the smaller event count of 31 to verify the trajectory is preserved.

The second limitation is the identifiability of the GAS persistence parameter ϕ at the boundary $\phi \to 1$. The posterior mean of $\phi = 0.92$ is high but not adjacent to the unit-root boundary; nonetheless, the small sample dimension ($T = 17$ years $\times J = 2$ technologies $= 34$ cells) provides limited information for distinguishing $\phi \in (0.85, 0.99)$ from $\phi = 1$. The posterior credible interval $[0.78, 0.99]$ contains values for which the trajectory would be near-unit-root and the long-run mean parameter ω would be weakly identified. The Bayesian inference accommodates this boundary identification through the prior structure, but for frequentist applications the boundary case would require explicit accommodation through a near-unit-root inference procedure.

The third limitation is the sample-composition heterogeneity across regions. The sample is dominated by Europe-EU-27 projects (62% of the sample) and the EUA-price-mediated mechanism is mechanically operative only for European projects. The region-indicator covariates absorb the within-region heterogeneity in baseline hazard, but the interaction term $\mathrm{Blue}_i \cdot z_t$ is identified primarily off the European subsample, with North American and Asia-Pacific projects entering the identification only through the assumption that the EUA-price-mediated mechanism is universal in its sign and approximate magnitude. The thesis precursor analyses (**(author?)**, **(year?)**, Chapter 7 "North America-only TVP analysis") document that the trajectory shape is preserved in the

North American subsample, supporting the universality interpretation; nonetheless, a fully region-stratified GAS-TVP specification would be a stronger identification design and is left to future work.

6.4 Generalisability of the methodology

The score-driven TVP approach we develop is a general-purpose tool for survival modelling under sparse-event conditions and suspected parameter instability. Three application domains for which the methodology should be useful include: (i) early-stage venture failure analysis, in which firm-level survival is governed by sector-specific time-varying hazard structures; (ii) project-finance termination, in which infrastructure-project failure may depend on time-varying macroeconomic conditions; and (iii) clinical-trial dropout, in which adherence may depend on time-varying treatment-protocol revisions. The required preconditions are: a binary failure outcome observable at a discrete frequency; a parameter of interest whose time-variation is plausibly smooth rather than abrupt; and a sample window of sufficient length to support the GAS recursion (we recommend $T \geq 15$ periods at minimum for a stable persistence parameter).

Beyond these three illustrative domains, the methodology should generalise to any setting in which (a) a small number of binary failure events are observed across a discrete time grid, (b) a covariate-conditional parameter of interest is suspected to vary over time in response to slowly evolving structural or policy conditions, and (c) the sample window contains at least one identified regime transition that constant-parameter alternatives cannot accommodate without functional-form misspecification. Concrete examples that satisfy these conditions and would benefit from the GAS-TVP approach include: bank-failure modelling under regulatory-regime shifts (Basel III phase-in, post-2008 stress-testing rollout), sovereign-default modelling under fiscal-rule revisions, climate-litigation-outcome modelling under judicial-doctrine evolution, and patent-survival modelling under examiner-policy revisions. In each case, the constant-parameter alternative will tend to either average across regimes (producing attenuated coefficient estimates) or absorb the regime variation into a poorly-identified block-step pattern (producing the same kind of artefact we document in the M2 specification of Section 5.1).

Two limitations of the cross-domain generalisability warrant explicit acknowledgement. First, the GAS recursion assumes a single dominant source of time-variation in the parameter of interest; settings with multiple orthogonal time-varying mechanisms operating simultaneously would require a multi-dimensional score-driven extension, the methodology for which is well-developed in the GAS literature [14] but not exercised in the present paper. Second, the rolling-one-step-ahead validation design that we use as the primary out-of-sample test is most powerful when the time-variation is monotonic or near-monotonic; oscillatory time-variation (e.g., business-cycle-conditional hazard differentials) would require alternative validation designs that condition on the cycle phase. The research agenda implied by these limitations — multivariate score-driven survival models with cycle-conditional validation — is left to follow-up work. The present paper establishes the univariate-score, monotonic-time-variation case in self-contained form, on which the multivariate and cycle-conditional extensions can build.

7 Conclusion

This paper has developed a time-varying-parameter state-space approach to survival modelling under sparse-event data conditions, in which the conditional hazard depends on a generative-process parameter whose evolution is driven by the score of the predictive likelihood. The methodological contribution is the principled combination of the Generalized Autoregressive Score (GAS) framework of [7] with the Bayesian state-space inference machinery of [11], applied to a complementary log-log hazard model with a single time-varying interaction coefficient. The OOS-comparison framework we apply — three independent test designs combined with the Diebold-Mariano-Hansen-Lunde-Nason inferential procedure — is, to our knowledge, the first formal application of the DM-MCS apparatus to survival modelling under sparse events.

The principal inferential conclusion is the dominance of the score-driven specification (M3) over the constant-parameter (M1) and parameter-driven block-transition (M2) alternatives. Under the rolling-one-step-ahead test design, M3 is the unique element of the Hansen-Lunde-Nason Model Confidence Set at $\alpha = 0.10$; the DM-HLN test statistics for M3 versus M1 and M3 versus M2 are $+5.59$ and $+4.80$ respectively ($p < 0.0001$). The advantage is concentrated in the 2023 cancellation wave, where the score-driven recursion correctly anticipates the late-sample intensification of the failure process by exploiting persistence from the preceding EUA-price-rise period.

The substantive application — the carbon-conditional implementation risk of hydrogen projects — demonstrates the empirical relevance of the methodology. The estimated trajectory of $\beta_{\text{int}}(t)$ intensifies monotonically from -0.46 in 2010 to -1.49 in 2024, with a discernible structural break around 2020 corresponding to the European Green Deal announcement and the subsequent Market Stability Reserve recalibration. The trajectory is consistent with the real-options interpretation in which the differential carbon payoff of clean over fossil hydrogen has risen with the EUA price, raising the optimal cancellation threshold for blue projects relative to green.

Three limitations are explicitly acknowledged: the event-timing uncertainty in the on-hold component of the failure outcome, the boundary identifiability of the GAS persistence parameter under small sample sizes, and the region-composition heterogeneity in the identification of the interaction effect. Each limitation is addressable in extensions to the present analysis, and the cancellation-specific subset of the failure outcome is robust to all three.

The methodology is generalisable to other survival applications with sparse-event data, suspected parameter instability, and a discrete temporal frequency. We anticipate that the score-driven TVP framework will be useful in venture-failure analysis, project-finance termination, and clinical-trial dropout modelling, among other domains. The replication package, including the full HMC implementation, the DM-test routines, and the rolling-window cross-validation framework, is available at `[github.com/SakeSaak/paper1_tvp]`.

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